

PVsyst - Simulation report

Standalone system

Project: off grid

Variant: New simulation variant

Standalone system with batteries

System power: 11.16 kWp

المنيا - Egypt



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PVsyst V8.1.3

VC0, Simulation date:
06/06/26 04:06
with V8.1.3

Project summary

Geographical Site

المنيا
Egypt

Situation

Latitude 30.8933 °(N)
Longitude 31.4330 °(E)
Altitude 30 m
Time zone UTC+2

Project settings

Albedo 0.20

Weather data

المنيا
Meteonorm 9.0 station (modified by user) - Synthetic

System summary

Standalone system

Orientation #1

Fixed plane

Tilt/Azimuth 26 / 0 °

System information

PV Array

Nb. of modules 18 units
Pnom total 11.16 kWp

Standalone system with batteries

User's needs

Daily household consumers
Constant over the year
Average 36.6 kWh/Day

Battery pack

Technology Lithium-ion, LFP
Nb. of units 9 units
Voltage 48 V
Capacity 911 Ah

Results summary

Simulation step	Hourly				
Useful energy from solar	13191 kWh/year	Specific production	1182 kWh/kWp/year	Perf. Ratio PR	54.03 %
Missing Energy	159 kWh/year	Available solar energy	21110 kWh/year	Solar Fraction SF	98.73 %
Excess (unused)	7334 kWh/year				

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General parameters

Standalone system

Simulation step

Hourly

Orientation #1

Fixed plane

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Standalone system with batteries

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

User's needs

Daily household consumers
Constant over the year
Average 36.6 kWh/Day

PV Array Characteristics

PV module

Manufacturer Generic
Model JKM-66HL4M-BDV-620
(Original PVsyst database)
Unit Nom. Power 620 Wp
Number of PV modules 18 units
Nominal (STC) 11.16 kWp
Modules 9 units x 2 In series

At operating cond. (50°C)

Pmpp 10.35 kWp
U mpp 75 V
I mpp 138 A

Total PV power

Nominal (STC) 11 kWp
Total 18 modules
Module area 48.6 m²
Cell area 45.4 m²

Controller

Universal controller
Technology MPPT converter
Temp coeff. -5.0 mV/°C/Elem.

Converter

Maxi and EURO efficiencies 98.0 / 96.5 %

Battery Management control

Threshold commands as SOC calculation
Charging SOC = 0.96 / 0.80
Discharging SOC = 0.10 / 0.35

Battery Storage

Battery

Manufacturer Generic
Model US5000B_100Ah
Technology Lithium-ion, LFP
Nb. of units 9 in parallel
Discharging min. SOC 10.0 %
Stored energy 39.3 kWh

Battery Pack Characteristics

Voltage 48 V
Nominal Capacity 911 Ah (C10)
Temperature Fixed 20 °C

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 8.9 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.75 %

Module mismatch losses

Loss Fraction 0.50 % at MPP

Strings Mismatch loss

Loss Fraction 0.10 %



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Array losses

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, $n(\text{glass})=1.526$, $n(\text{AR})=1.290$

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000



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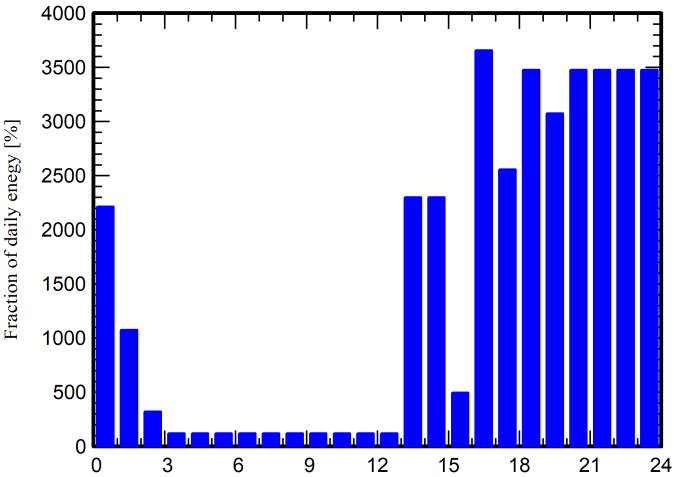
Detailed User's needs

Daily household consumers, Constant over the year, average = 36.6 kWh/day

Annual values

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
hood	1	55/lamp	1.0	55
lambs/ fan	1	755/app	8.5	6418
tv / labtop/ charger/	1	405/app	8.0	3240
router / Fridge	1		24	2820
Washing machines / pump	1		3	4020
air conditioning	1	2200 tot	7.5	16500
heater/ hair heater	1	3500 tot	1.0	3500
Stand-by consumers			24.0	24
Total daily energy				36577

Hourly distribution





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Main results

Simulation step

Hourly

System Production

Useful energy from solar 13191 kWh/year
Available solar energy 21110 kWh/year
Excess (unused) 7334 kWh/year

Perf. Ratio PR 54.03 %
Solar Fraction SF 98.73 %

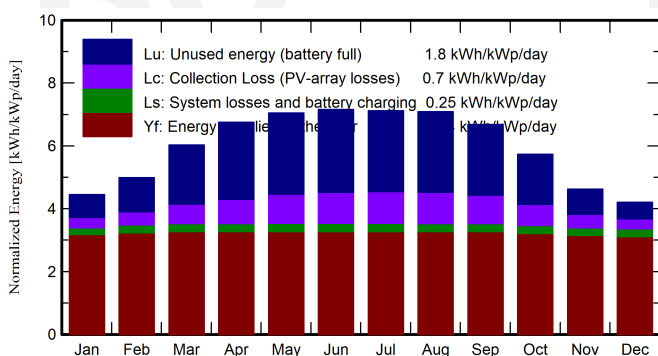
Loss of Load

Time Fraction 0.6 %
Missing Energy 159 kWh/year

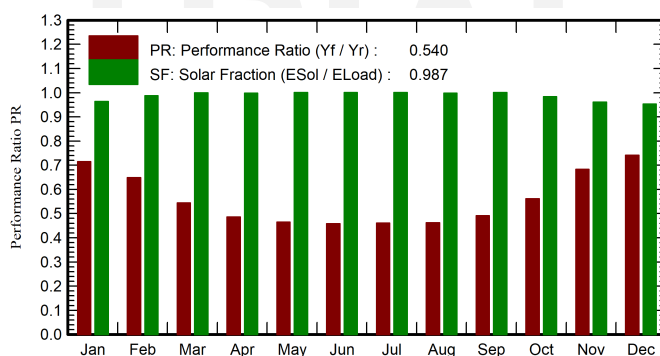
Battery ageing (State of Wear)

Cycles SOW 94.7 %
Static SOW 93.3 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	GlobEff	E_Avail	EUnused	E_Miss	E_User	E_Load	SolFrac
	kWh/m ²	kWh/m ²	kWh	kWh	kWh	kWh	kWh	ratio
January	98.3	135.5	1393	250	31.4	1102	1134	0.964
February	109.8	137.3	1393	335	13.4	1011	1024	0.987
March	162.5	183.1	1829	645	0.0	1134	1134	1.000
April	193.4	198.5	1962	817	0.0	1097	1097	0.998
May	227.7	213.5	2078	893	0.0	1134	1134	1.001
June	234.1	209.5	2024	878	0.0	1097	1097	1.000
July	235.8	215.2	2071	887	0.0	1134	1134	1.001
August	217.2	214.8	2065	883	0.0	1134	1134	0.999
September	180.3	196.3	1897	748	0.0	1097	1097	1.000
October	143.2	174.5	1717	551	19.4	1115	1134	0.983
November	102.3	136.4	1370	266	40.8	1057	1097	0.961
December	90.8	128.3	1312	181	54.2	1080	1134	0.953
Year	1995.3	2142.8	21110	7334	159.1	13191	13350	0.987

Legends

GlobHor Global horizontal irradiation
GlobEff Effective Global, corr. for IAM and shadings
E_Avail Available Solar Energy
EUnused Unused energy (battery full)
E_Miss Missing energy

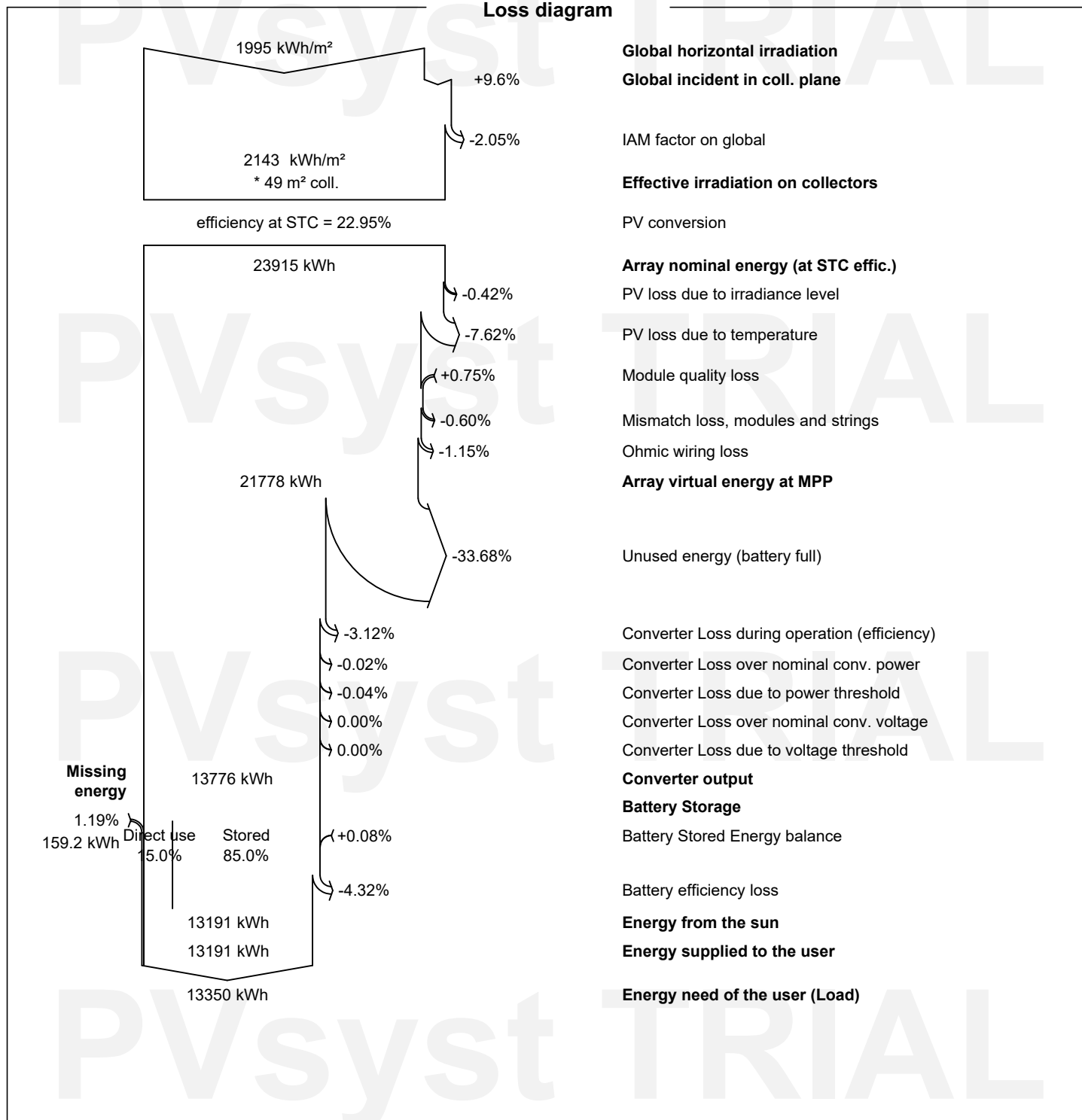
E_User Energy supplied to the user
E_Load Energy need of the user (Load)
SolFrac Solar fraction (EUsed / ELoad)



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Loss diagram





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Predef. graphs

Daily Input/Output diagram

